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Modelling and simulating multi-echelon food systems

Jack G.A.J. van der Vorst ^{a,*}, Adrie J.M. Beulens ^{b,1}, Paul van Beek ^a

^a Wageningen Agricultural University, Sub-department of Management Studies, Hollandseweg 1, 6706 KN Wageningen, The Netherlands

^b Wageningen Agricultural University, Sub-department of Information Technology, Dreijenplein 2, 6703 HB Wageningen, The Netherlands

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Abstract

This paper presents a method for modelling the dynamic behaviour of food supply chains and evaluating alternative designs of the supply chain by applying discrete-event simulation. The modelling method is based on the concepts of business processes, design variables at strategic and operational levels, performance indicators, and business entities, as these concepts allow relevant aspects of a food supply chain to be described and modelled. Our simulation model is based on timed coloured Petri-nets to support decision-making when redesigning a supply chain for chilled food products. Scenario analysis allowed us to evaluate the potential benefits of alternative designs for this supply chain in terms of logistical performance. In this case study, major benefits were identified. The results of the scenario analysis are compared with the outcomes of a pilot project where one scenario was implemented in real life. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Supply chain modelling; Discrete-event simulation; Modelling method

1. Introduction

Supply Chain Management (SCM) has generated many research efforts in recent years because of the perception that actions of one actor in the chain can influence the cost and value added by all other actors in the chain. In recent literature on

SCM one identifies supply chains (Fig. 1) as entities that need to be managed as a whole in order to achieve better customer value and more revenue at lower overall costs while satisfying a variety of constraints (see Beers et al., 1998). In our view, chain performance improvement is impeded by the presence of uncertainties to do with decision-making. The uncertainty observed daily, such as late deliveries, machine breakdowns, order cancellations, leads to increased inventories (Davis, 1993), additional capacities or unnecessary slack time. The uncertainty that plagues complex networks is a real problem, as is finding remedies to reduce that uncertainty. SCM offers the opportunity to improve

* Corresponding author. Tel.: +31-317-48-3644/4160; fax: +31-317-48-4763.

E-mail address: jack.vandervorst@alg.bk.wau.nl (J.G.A.J. van der Vorst).

¹ Fax: +31-317-483158.

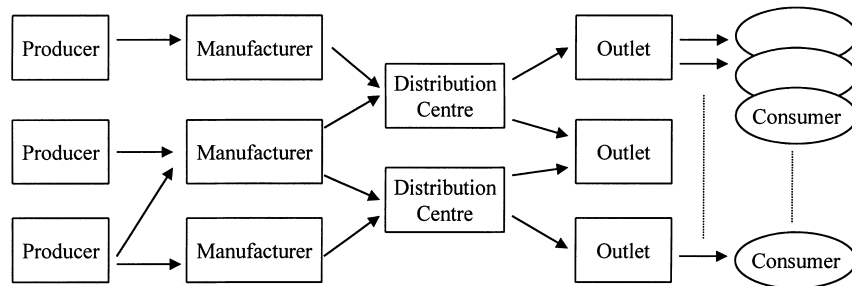


Fig. 1. Schematic diagram of a generic food supply chain.

chain performances by reducing these uncertainties. It is clear that there is a need for some level of co-ordination of activities and processes within and between organisations in the supply chain to reduce uncertainties and add more value for customers (Cooper et al., 1997). The key attributes in this process seem to be the elimination, synchronisation or reduction of time-consuming processes and the availability of information (see also Towill et al., 1992; Lee et al., 1997). Van der Vorst and Beulens (1999) present a research method to formulate relevant chain redesigns in which the identification of uncertainties in supply chain decision processes plays a crucial role. This paper presents a method for modelling the dynamic behaviour of food supply chains and evaluating alternative designs of the supply chain infrastructure and operational management and control by applying discrete-event simulation.

1.1. Literature review

Jones and Riley (1985) claim that SCM should fulfil final customer service requirements, define where to position inventories along the supply chain and how much to stock at each point, and it should develop appropriate policies and procedures for managing a supply chain as a single entity. This is in contrast with managing one specific role within a supply chain (for example manufacturing or warehousing); a situation in which each role is basically only concerned with its direct customers and suppliers. In an extended literature review Bechtel and Jayaram (1997) showed that

the term SCM is often misused and is variously defined. Christopher (1992) gives a suitable definition of a supply chain from a logistical point of view. He states that a supply chain is “a network of organisations that are involved, through upstream and downstream linkages, in different processes and activities that produce value in the form of products and services in the hands of the ultimate consumer”.

In recent years, Operational Researchers have put more emphasis on modelling and analysing multi-echelon systems. Though SCM is relatively new, the idea of co-ordinated planning is not. Multi-echelon inventory/distribution systems were first studied nearly 40 years ago (Clark and Scarf, 1960). Since then, many Operations Researchers have investigated multi-echelon inventory and distribution systems. Thomas and Griffin (1996) give an extensive review, addressing co-ordinated planning between two or more stages of the supply chain and placing particular emphasis on models appropriate for modelling the chain. It is difficult to find algorithms and models for inventory policies in multi-echelon systems with stochastic demand and batch or periodic ordering. Many authors consider a two-echelon distribution system with identical retail sites and Poisson demand, and then develop an approximate model of system costs or performance as a function of inventory levels; simulation is used to evaluate the model (Graves, 1996). Jackson (1988) and Schwarz (1989) provide excellent reviews of this literature. Lee and Billington (1992) provide an extensive discussion of the dangers of poorly managed supply chain inventories. Furthermore, they state

that supply chain analysis is much more than inventory modelling. It can be extended to distribution strategy analysis and to other types of supply chain problems (Lee and Billington, 1995). Van Houtum et al. (1996) review stochastic multi-echelon planning and control systems for production and distribution-to-stock environments. Diks et al. (1996) state that one of the main difficulties in cost-efficient and effective SCM is to determine the target service levels (associated with the service measures selected), so that pre-specified external service targets are met at lowest cost. They give an extended review of contributions in controlling divergent multi-echelon systems with service level constraints, considering costs of holding inventory, order costs and the cost of running out of stock.

1.2. Research objectives

The main drawback of most analytical models is the fact that numerous constraints have to be satisfied before results can be applied in practice. Most models only take few variables into account, for example, inventory and cost of running out of stock, and ignore other cost such as order processing, handling and transportation. Furthermore, these models neglect capacity restrictions and treat systems as not correlated in time, ignoring the complex material relationships that occur if upstream installations fail to serve downstream installations (see Van Houtum et al., 1996), and this is the essence of SCM. SCM specifically focuses on the interrelated two-way flow of products (materials and services) and information with the associated managerial and operational activities to obtain a high degree of flexibility (responsiveness).

Silver et al. (1998) state that if quantitative models are to be more useful as aids for managerial decision making, they must be more realistic representations of the problem; in particular, they must permit some of the usual ‘givens’ to be treated as decision variables. Whybark (1988) expressed the same viewpoint when he stated “in our classes we still approach the problems from the standpoint of ‘given the following constraints,

what is the optimum expenditure to achieve maximum profits?” In practice, the reverse occurs: ‘how can we redefine these constraints in order to improve our performance?’” Therefore, instead of drawing up our own constraints for the supply chain problem, we decided to let those with the problem, determine the constraints to the problem being investigated. To achieve this, we depart from chain objectives and identify performance indicators and variables that influence logistical chain performance in consult with those with first-hand knowledge of the problem. This yields the constraints to be set on the modelling process. The research reported here thus aimed at improving chain performance by tracing and evaluating alternative designs of the supply chain infrastructure and operational management and control.

A quantitative analysis of multi-echelon systems enables the performance of a supply chain to be predicted. Mathematical approaches require too many simplifications to model realistic chain problems. Discrete event simulation allows logistic systems to be modelled more realistic. This paper therefore reports the findings of our research into the discrete-event modelling and simulation of alternative designs for a supply chain for chilled salads. Our simulation model can be compared with a multi-echelon, multi-product system with periodic ordering, which evaluates alternative chain designs with respect to total chain cost and a number of service elements.

In Section 2, we will introduce the fundamental components of our modelling approach. Section 3 describes our case study in the food industry, which shows the efficacy of our modelling approach. In Section 4 we will present our conclusions and indicate future research directions.

2. Modelling and redesigning a supply chain

In this section, we will first discuss the fundamental modelling components to model the dynamic behaviour of food supply chains. These are business processes, performance indicators, design variables and business entities. Second, we will discuss the characteristics of discrete-event simulation and, consecutively, focus on Petri-net

modelling. We will end this section with a discussion of supply chain redesign.

2.1. The components for modelling a supply chain

2.1.1. Business processes

The supply chain comprises a set of business processes and their interrelationships. Fig. 2 depicts some of the processes that can be found in the generic food supply chain presented in Fig. 1. Taking a process approach implies adopting the customer's point of view. As Davenport puts it: "Processes are the structure by which an organisation does what is necessary to produce value for its customers", and a process is a structured, measured set of activities designed to produce a specified output for a particular customer or market (Davenport, 1993). Key differences between the traditional functional approach and the process approach are that the focus of every process is on meeting final customer requirements and that the firm is organised around these processes. Hence, the identification of value-adding processes in the supply chain (focusing on the final customer service requirements) and their mutual relationships is crucial. A process comprises a set of tasks to be executed in a specific order and according to a specific policy. Each process is an elementary system in the order life cycle. Examples are order generation, production, order picking, stock holding, etc. We distinguish between production processes (transforming goods) and administrative and decision processes (transforming information). How these processes are organised is mainly

determined by the time frames applied. A process can be executed during a certain time window; it has a starting time, a process throughput time (that depends on the work to be done and the available capacity) and a time at which it should be finished.

2.1.2. Design variables

Design variables refer to the management and control of a business process. It concerns variables that determine the place, policy and/or procedure, timing, and the information and organisational infrastructures to be used when executing the tasks of a process. The total setting of all design variables in the supply chain determines the overall chain design. SCM offers the opportunity to increase information transparency, co-ordinate decision policies and shorten chain throughput times to increase responsiveness to changing consumer demands. We distinguish between two categories of design variables.

1. *Design variables at configuration level.* Variables at this level determine the configuration of the supply chain with regard to parties involved, roles to be performed, manner of co-operation, constraints to executing roles, and the IT and physical infrastructures to be used. The result will be a known set of participants with specified roles in the chain and required infrastructures.
2. *Design variables at operational management and control level.* At this level we need to determine how the co-operation and integrated planning of operations can be improved within the given configuration, in terms of the timing, accuracy, quality, quantity, etc. of goods and information flows and of business processes.

When logistical chain performance does not satisfy chain objectives, numerous alternative designs can be realised by adjusting a selection of design variables at either or both of these control levels. In the short-term, design variables at the operational management and control level are the prime candidates for changing short-term 'givens' (Silver, 1991): for example, shortening of the lead time. For more drastic results in the longer term, design variables at configuration level should be identified and changed: for example, selecting another supplier.

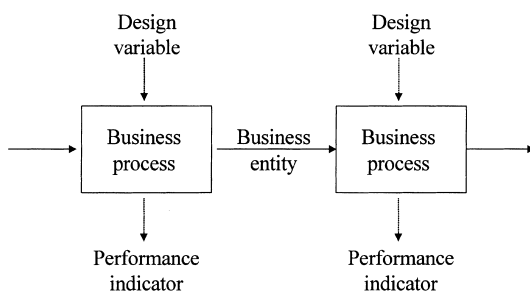


Fig. 2. The components for modelling a supply chain.

2.1.3. Performance indicators

Firms increasingly think in terms of competing as part of a supply chain against other supply chains, rather than as a single firm against other individual firms. Hence, main performance requirements are defined by chain customer service requirements. In food supply chains, consumer service requirements especially refer to product availability (no out of stocks and large assortment), product quality (particularly with respect to remaining shelf life, health and tracing) and acceptable price. The various actors in the total supply chain determine what levels of customer service the supply chain should reach. Consequently, performance requirements for each role and for each process in the supply chain can be derived depending on the architecture of the chain, which leads to a list of relevant performance indicators. A performance indicator is defined as a variable indicating the effectiveness and/or efficiency of a part of or the whole of the processes or systems compared to a given norm/target or plan (Fortuin, 1988). If realised values for these performance indicators differ from set-point values, chain redesign will be inevitable.

2.1.4. Business entities

We have previously described supply chains as networks of organisations that are involved, through upstream and downstream linkages, in different processes and activities that produce value in the form of products and services in the hands of the ultimate consumer. The linkages between processes encompass goods and information flows (we will now disregard financial flows). The flows comprise sets of products or information that are transformed by place, time or state. A business entity is used to represent an information flow and/or goods flow, or it may represent the availability of certain capacities of resources (machines, labour). It is therefore a complex, well-identified entity with a set of descriptive attributes. Each business entity has:

- a unique identification;
- a time stamp indicating the time at which it is released for further processing;
- descriptive attributes;

- a number of characteristic entities (a repeating group) that depend on the business entity for their existence;
- attributes that relate the business entity and/or the characteristic entities to attributes of the business entities used in processing the business entity.

For example, a shipping list can be identified by shipment number 7, released on Monday at 10 o'clock in week 12, is shipped from distribution centre 3 and has a delivery for retail outlet 11. The list further contains a number of shipment lines; each characteristic entity describes one part of the delivery for a product from a particular batch (attributes of such a line are batch number and quantity delivered). The batch number refers to a combination of a product number and the remaining shelf life, and is an example of a relationship of that entity to entities used to produce it. This is necessary for tracing products and for calculating performance indicators. When modelling a supply chain we see a continuous process of the creation and flow of transitions of these business entities, which means a conversion of place, time and state/content.

2.2. Discrete-event simulation

A supply chain model consists of the instantiations of the modelling components; the relevant business processes of all parties in the supply chain, design variables at both the configuration and the operational management level (e.g. all policies and precedence relations in time) and relevant performance indicators. The model should be capable of calculating the state, time and place of each business entity after each transition, and the performance of the chain. Transition systems are characterised by the fact that the system has a discrete array of possible states and there is a sequence of events that moves the system from one state to another (Van Hee, 1994). Systems like this are also referred to as Discrete Event Dynamic Systems. The dynamic character of the model is expressed in a number of equations, which calculate values for local model variables during a process. For example, when orders are processed

into order pick lists, inventory levels are checked and compared with total demand. If total demand at that time exceeds total inventory per product, shortages may be divided according to a rationing policy.

Each process is related to an actor in the supply chain where it is executed using and processing complex business entities, which enter the process and satisfy certain time window constraints. Processes are triggered by events in the network such as the arrival of a business entity (the output of a former transition) or system time reaching a pre-set time stamp (e.g., Tuesday 09.15 h). The timing of each process is crucial. We model the supply chain as a network of administrative and physical logistical activities with precedence relations in time. The business entity that is the output of one transition forms the input for the next transition. In this way the timing of each transition (logistical process) can be arranged, and throughput times of processes and waiting times can be calculated. There is a schedule of pre-set times at which processes (such as ordering, inventory control, receiving goods, and processing write-offs) may start, and must be completed (that is the time at which the business entities produced must be dispatched to receiving processes). Further, we may distinguish fixed and variable processing times per business process, depending on the content of the entities to be processed (for example the number of products to be picked). Processes can have preconditions, i.e., will not be instantly triggered by an event, unless certain restrictions are fulfilled. In this way the model approaches reality. For example, order picking does not start when picking orders arrive, unless capacity is available.

Using time stamps and/or batch numbers as attribute values in business entities, we keep track of the connection between input and processed entities for tracking and tracing and performance measurement. Given the importance of remaining product shelf lives, it should be possible to calculate these values for all products at all times. The value of the attributes influences the processes in which they are transformed. For example, products with the shortest remaining shelf life should be picked and delivered first, unless their remain-

ing shelf life has diminished to zero, in which case they should be discarded. These processes result in complex transitions of business entities, which are further complicated by the existence of combined truck loads. For example, a shipment comprises multiple products and can also encompass multiple customers.

The dynamics and complexity of these processes and the fact that we are focusing on numerous performance indicators reduce the applicability of analytical models for modelling complete supply chains. However, these analytical models can be used to determine lower and/or upper boundaries for instances of design variables.

2.3. *Petri-net modelling*

In our research we work with ExSpect (Van Hee et al., 1989), a specification language based on timed coloured Petri-nets, to model and simulate the dynamic behaviour of the supply chain. We opted for Petri-nets because these allow us to realistically represent the problem applying our modelling concepts. With Petri-nets, a system can be modelled as a hierarchical network of business processes; a system is an aggregate of places, transitions and possible subsystems. In this way the system, the supply chain, can be divided into linkages with logistical activities and each activity can be divided into successive process steps. A business entity is represented in the model by a token with a complex structure (or 'colour'). The colours used in our modelling approach match the definition of business entity. Each activity is related with all other activities by the flow of business entities (tokens) through the model. Timed Petri-nets use a timing mechanism that associates time with tokens and specifies delays by an interval. The state of a Petri-net is related to the presence of tokens in places, and is only defined at points in time when events occur. Petri-nets are capable of handling parallelism and synchronisation and result in simulation structures that are logistically transparent. For an elaboration on timed coloured Petri-nets see Van der Aalst (1992) and Van der Zee (1997).

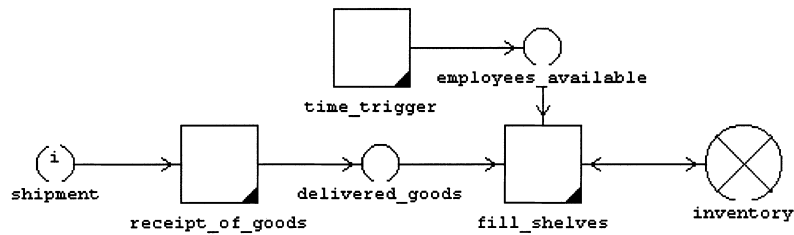


Fig. 3. Example of a timed coloured Petri-net model in ExSpect (a square represents a business process, a circle a path for business entities and a circle with a cross in it a store/database).

In order to formulate Petri-nets there must be a descriptive model of the system. We have found that the Organisation Description Language (ODL), focusing on our modelling concepts, can be used to do this (Uijttenbroek et al., 1995). ODL is capable of describing complex business entities and their interrelationships. It focuses on the inputs and outputs and particularly on successive steps of transitions. Because of the clear and detailed description of the system, ODL allows experts and users to validate the descriptive models.

We will give a small *example* of a Petri-net model representing part of a food supply chain (Fig. 3). It concerns the delivery of goods to a retail outlet and, consecutively, the upgrading of retail stock levels. The entity shipment $S(SN, W, R, T1, RG(BN, SQ))$ refers to a shipment with unique number SN that arrives from distribution centre W at retail outlet R at time $T1$, with a Repeating Group (RG) of a shipped quantity SQ for each unique (underlined) batch number BN . The batch number refers to a combination of a product number and a remaining shelf life. The arrival of this entity triggers the business process 'goods receipt', which transforms *shipment* in the business entity *delivered_goods* $DG(DGN, T2, RG(BN', DQ))$. At time $T3$ the *employees* who fill the shelves in the outlet become available and start filling the shelves at a certain speed. Once all goods have been put on the shelves (after a certain process throughput time DT), *inventory* $I(IN, L, T, RG(BN, IQ))$ is upgraded to $I(IN, L, T3 + DT, RG(BN', IQ'))$. This denotes an inventory record with number IN for link L , recorded at time stamp $T3 + DT$, and for each batch number BN inventory quantity IQ .

2.4. Supply chain redesign

This paper focuses on the modelling and analysis of alternative supply chain designs. Elsewhere (Van der Vorst and Beulens, 1999), we give a detailed description of a plan of approach to arrive at supply chain scenarios. It has two main components to identify relevant chain redesigns. First, a conceptual logistical framework for chain analysis to identify relevant redesign variables. Second, the identification of sources of decision-making uncertainties in the supply chain to define relevant scenarios. In accordance with Davis (1993), we state that a company can make great strides by understanding the relative impact of different sources of uncertainty in the system and by then working to reduce (or avoid) their impact. The sources of uncertainty indicate the opportunities for SCM and help identify and rank relevant redesign variables in the supply chain. The assessment of the estimated impact of each source of uncertainty on defined performance indicators results in a ranked list of sources of uncertainty. Via cause-effect analysis the relevant redesign variables that are likely to influence the main sources of uncertainty are derived and transformed into scenarios. The main sources of uncertainty we identified in food chains are the length of the order forecast horizon, information availability and data timeliness, decision policies used, and inherent supply, process and demand uncertainties in the supply chain. The approach has led to a general list of redesign principles in supply chain perspective that impact supply chain performance (cost and service) and includes the following:

- diminish time windows for business processes (waiting times, processing times);
- synchronise, eliminate or reallocate business processes;
- shorten cycle times and increase the execution frequency of business processes;
- co-ordinate decision policies;
- implement real-time information systems;
- standardise and implement definitions of supply chain metrics;
- create information transparency, for example of consumer demand information or production schedules.

3. Case study

3.1. The supply chain for chilled salads

We applied the research approach and modelling method in an actual food supply chain for chilled salads in the Netherlands, comprising a producer and a retailer with one distribution centre (DC) which supplies exactly 100 retail outlets. These organisations started a chain partnership to improve business performance to the final consumer via more responsive operations combined with better use of resources. In this partnership, emphasis was on the distribution of chilled salads. The general objective was to improve customer service at lower total chain costs, as this should boost sales and profits for all chain participants. More specifically, the producer wanted to create more space in retail outlets to expand his product assortment and improve product freshness for consumers. To measure the effectiveness and efficiency of alternative chain designs, we translated this objective into a number of performance indi-

cators for this supply chain, listed in Table 1. The cost drivers were determined in discussions with the participating organisations. Note that the outcomes of such discussions may differ for other supply chains, which may have great impact on benefits to be obtained.

Fig. 4 depicts the main modelled business processes in the supply chain for chilled salads, based on a customer order life cycle analysis. In the retail outlet salads are sold during opening hours, whilst inventory levels are adjusted at the end of the day. Twice a week retail outlets place orders at the DC according to a base stock policy (basically orders are placed to replenish available shelf space). Because of a one-day lead time, backordering does not apply. The order batch quantity is a fixed number of boxes of consumer products that can vary for each product. During closing hours, all stocked goods are checked for their remaining shelf life. If there is none, goods are written off.

At the DC the outlet orders are processed at fixed times, which results in order pick lists per retail outlet. Some time later orders for all products are picked and collected into shipments, which, after a short delay, are shipped to outlets. The DC supplies retail outlets with chilled salads three times a week from stock. This is done simultaneously with other fresh products, according to a fixed delivery schedule with a lead-time of one day. The times at which corresponding shipments leave the DC are pre-set, but the actual arrival time at the site depends on process throughput times. If there are shortages, a rationing policy is followed. The DC generates an order twice a week based on sales forecasts (derived from outlet orders and historical sales patterns), a safety factor and the previous day's inventory levels. The producer processes the orders at fixed times and

Table 1
Performance indicators in the supply chain for chilled salads

Supply chain costs	Service
Holding costs at DC and retail outlets	Number of stock-outs per hour
Processing costs related to all logistical processes in the supply chain	Delivery reliability of producer and DC per produce
Costs of product write-offs and necessary price reductions	Average remaining product freshness
	Utilisation degrees of transport carriers
	Product assortment

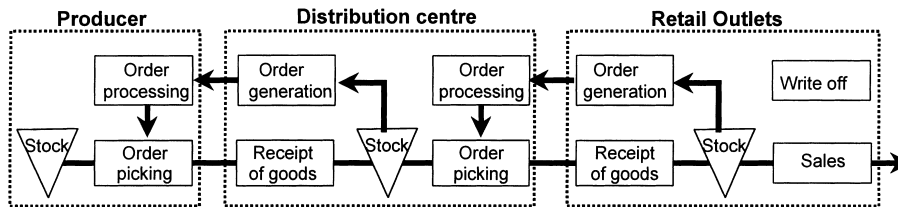


Fig. 4. The business processes modelled within the supply chain for chilled salads.

supplies the DC with 63 different product types with an order lead-time of three days. Also in this case, a rationing policy is used if there are shortages. Since sales to this particular retailer involve only a marginal proportion of the producer's total sales, it is unlikely that a redesign of this chain will cause the producer's inventory levels to change substantially. For this reason, we did not measure the producer's inventory levels in our study.

3.2. Supply chain redesign

The objective of the simulation study was to model the dynamic behaviour of the supply chain and to evaluate the consequences of different configurations on indicators of logistical performance. Objectives and performance indicators of relevant processes were defined, along with relationships between processes; for the latter we focused on the timing and duration of processes, the information required to complete the task and the uncertainty related to that particular task. A number of *design variables* seemed relevant in the process of redesigning the supply chain (Table 2). Of these, the reduction of time windows showed to

be most important, since it increases responsiveness and reduces the order and planning forecast horizons. Furthermore, decision-making processes can be improved by increasing the information availability and timeliness, by decreasing human influence or by shifting decisions to other chain participants.

3.3. The simulation model

The simulation model should be capable of calculating the values for all performance indicators (see Table 1) for alternative supply chain designs (see Table 2). Our model comprises two retail outlets. One is an average retail outlet in which relevant performance indicators are measured based on the input data for that outlet (in particular, shelf space, consumer demand and order batch sizes for each product, and process time windows and fixed and variable cost for each business process). The other is a huge outlet that represents total consumer demand per product of all other outlets to simulate the total volume of goods and information flows in the supply chain. This is required in order to satisfy capacity con-

Table 2
Main redesign principles for the chilled salads supply chain

Redesign level	Redesign principles
Configuration	Implementing a Computer Assisted Ordering (CAO) system in retail outlets (order policy based on forecasts), a real-time inventory system at the retail DC, and EDI with the producer
Operational management and control	Decreasing the lead-time from three days to less, including changing the time windows of each critical business process Increasing ordering and delivery frequencies in the supply chain; between producer and DC from 2 to 3, 4 or 5 times a week and between DC to outlets from 3 to 4, 5 or 6 times a week Allocating the DC ordering policy at the producer; simply passing on aggregates of the outlets' orders and current inventory levels

straints and calculate utilisation degrees of transport carriers and trucks. Furthermore, it influences inventory levels at the DC and therefore inventory turnover and product freshness. The model specifically focuses on the results obtained in the average retail outlet and transforms these into generic results.

The model was fed with data on eight representative products to be extrapolated to the total assortment of goods. Each of these products represented a group of products with comparable consumer demand, available shelf space, order batch size and shelf life constraints. Total chain costs were calculated by extrapolating the results to the total product assortment.

The model was run with consumer demand data per product per hour for both outlets considered until an approximate steady state situation was accomplished. The data have been derived from data recorded by the point-of-sale systems in practice for 20 consecutive weeks. In order to be able to compare the simulation results of different scenarios on all performance indicators, scenarios were calibrated with respect to a 100% delivery performance to final consumers (thus 0% stock-outs). This means that each scenario was calibrated to minimum inventory levels in the retail outlet, while still satisfying consumer demand.

3.4. Simulation results

Numerous scenarios were simulated but we will focus on the most important results (only changes to the current situation are given). In all these scenarios EDI and a real-time inventory system were implemented in the supply chain (i.e., simulated by adjusting the duration and timeliness of correlated processes). This means that inventory levels were known at all times at the ordering process.

1. The producer's lead-time $L(p)$ decreased from three days to 24 hours and all related time windows of each critical process changed.
2. The order and delivery frequency between producer and DC $FR(p, dc)$ increased from 2 to 3, 4 or 5 times a week with $L(p) = 1$ day and $FR(d, co) = 3$.

3. The order and delivery frequency between DC and outlets $FR(d, co)$ increased from 3 to 4, 5 or 6 times a week with $L(p) = 1$ day and $FR(p, dc) = 3$.
4. The order and delivery frequency between producer and DC $FR(p, dc)$ increased to 4 and 5 respectively, with $L(p) = 1$ day and $FR(d, co) = 6$.
5. The base stock ordering policy in retail outlets changed into a system driven by order forecast (which enables several policies to be evaluated).

Total costs per link were calculated by adding costs related to all business processes in that link (including the costs of write-offs and price reductions). Total chain cost is the sum of the cost of producer and DC, and 100 times the costs made in the average retail outlet. Total process cost refers to total cost minus write-offs and price reductions. This gives an insight in the cost of human labour related to the execution of all business processes. The results are given in Table 3. Because of the confidentiality of the data we transformed all outcomes into changes relative to scenario 1b. For example the number 110 means a 10% increase compared with the value obtained in scenario 1b. Scenario 1a was the real-life scenario, whereas scenario 1b was a comparable scenario with $L(p) = 1$ day.

When $L(p)$ was decreased to one day, inventory levels at the DC decreased on average by 15% and throughput times fell by 14%. This resulted in fresher products delivered at retail outlets, leading to less write-offs. Inventory levels rose slightly in outlets because of the calibration process to obtain minimum inventory levels; it is not possible to install safety stock levels at which no out-of-stocks and no write-offs occur at these lower delivery frequencies. When inventory is too high, products perish because they remain on the shelf too long, and when inventory is too low, stocks run out. When the delivery frequency is increased, optimal inventory levels can be installed.

Table 3 shows that the most important results were obtained when $FR(p, dc)$ and $FR(d, co)$ were both increased by one (scenario 3a). Average chain stock levels fell by about 23% and chain throughput time fell by 15%, which means an average increase in remaining product shelf life of 2.2 days. Chain process cost increased by 4%, mainly

Table 3
Overview of the most important simulation results

	Scenario									
	1a	1b	2a	2b	2c	3a	3b	3c	4a	4b
<i>Performance indicator</i>										
$L(p)$	3	1	1	1	1	1	1	1	1	1
$FR(p, dc)$	2	2	3	4	5	3	3	3	4	5
$FR(d, co)$	3	3	3	3	3	4	5	6	6	6
<i>Producer</i>										
Total cost	101	100	103	105	108	103	103	102	104	106
<i>Distribution centre</i>										
Stock level	115	100	79	72	63	76	76	83	78	65
Throughput time	114	100	82	74	70	74	82	92	71	55
Total Cost	100	100	100	100	101	108	117	125	126	128
<i>Outlet</i>										
Stock level	94	100	100	100	100	77	74	78	77	80
Throughput time	98	100	105	108	104	91	87	86	91	88
Write-offs	110	100	42	35	29	32	29	32	35	29
Process cost	106	100	102	97	98	102	105	109	108	108
<i>Chain</i>										
Throughput time	114	100	96	95	92	85	85	89	84	77
Process cost	100	100	102	101	102	104	107	110	111	112
Total cost	103	100	86	88	88	87	87	90	95	91

because of higher order pick, transport and order costs. Total chain cost fell, as a result of fewer write-offs and price reductions. Furthermore, inventory reduction can be used for expansions in product assortment, which will generate more profit. Increasing the frequency still further resulted in relatively fewer improvements, since the current order batch size (on average, 12 consumer purchases) restricted better results; the turnover of most products was only three or four batches per week. If the order batch size was set to one individual consumer product, higher delivery frequencies became more beneficial. Simulation results show that this scenario will result in 26–38% additional improvements in inventory levels and product freshness. However, costs (especially order picking costs) may rise significantly (by about 23%).

It was clear that the implementation of a forecast-driven ordering system could further reduce inventory levels. The results of the simulation show an additional improvement in performance of 10–20% relative to the standard scenario. However, it is impossible to predict the exact benefits that would actually be obtained, because it

was impossible to simulate human interventions in the case of peak demand or other disturbances.

3.5. Model validation

We validated the simulation model and its results in four ways (cf. Law and Kelton, 1991). First, the model was logically validated on the basis of extensive ODL descriptions of the supply chain that were defined, corrected and approved by experts (chain participants). Second, in the modelling process the state of simulated sub-systems (the business entities) was printed out and compared with manual calculations to see if the program was operating as intended. Third, simulation of the current real life scenario (scenario 1a) revealed that performance metrics approached reality (empirical validation). However, inventory levels were slightly higher because stock-outs were eliminated in the simulation model (the model is calibrated to 100% delivery performance). Hence, remaining product shelf lives were slightly shorter. This is in accordance with the findings of Cachon

and Fisher (1997) who state that the elimination of final stock-outs requires an increase in inventory. Finally, we were able to obtain an indication of the validity of the results of the simulation study by implementing one scenario in real-life and measuring the performance changes. This also gave us insight into the organisational consequences of alternative chain designs, thereby revealing organisational constraints. The scenario implemented was one with $L(p) = 1$, $FR(p, dc) = 5$, $FR(w, dc) = 6$, an order-up-to ordering policy at the DC and an order forecast module at retail outlets to obtain maximum benefits. This scenario was implemented for six weeks in one typical retail outlet (the same one that was simulated) in which detailed information was gathered on relevant performance indicators. Target inventory levels were determined for all products based on peak weekly demand and a percentage of safety stock. The results are given in Table 4.

Table 4 shows that the model underestimated the benefits to be obtained for this particular scenario. The pilot showed higher benefits, for the following reasons:

- In the simulation study two ‘optimised’ scenarios (both 0% out-of-stock) were compared, which (virtually) increased inventories. In practice a ‘nearly optimised’ situation (we measured ca. 2.5% out-of-stock in retail outlets in the pilot) was compared with a situation with more slack (in general ca. 5% out-of-stock).
- The order forecast driven system in the outlet was very difficult to model.
- The simulation model was fed with 20 weeks of sales data, which showed more demand variability than sales data during the six weeks of the pilot study. This required the presence of higher safety stocks.
- In the pilot, the orders of the pilot outlet were given high priority in the DC in the event of

shortages (when possible, they received a 100% delivery) because people knew their performance was being measured.

Although the outcomes were not identical for the reasons given above, the experts concluded that the model did discriminate between different scenarios. They concluded the model did predict probable trends and directions in the benefits to be obtained. This was mainly because all participants had been involved in developing the simulation model and were fully aware of all underlying assumptions.

Which scenario, a supply chain should implement depends on the trade-off between service levels and total chain costs. From the pilot study it was clear that there are some organisational barriers to overcome before the suggested scenario could be implemented in real life. EDI and real-time control systems seem to be necessary together with training and education of personnel to make a reduction in lead-times and an increase of frequencies achievable.

4. Conclusions

Modelling the dynamic behaviour of supply chains is required to analyse the potential benefits of SCM. The method we have presented for modelling and evaluating alternative designs for multi-echelon food systems, focuses on modelling this dynamic behaviour without making too many concessions to the real problem. We opted for discrete event simulation to support supply chain decision-making, based on concepts of business process, design variables, performance indicators and business entity. When we tested the modelling method and concepts on a supply chain for chilled products, various major benefits were obtained. The dynamic behaviour of this supply chain was

Table 4
The results of the pilot project compared with the simulation study

Performance indicator	Pilot study	Simulation model
Inventory level DC	Decreased by 55%	Decreased by 41%
Inventory level retail outlet	Decreased by 38%	Decreased by 16%
Product freshness for consumer	Increased by ca. 5.0 days	Increased by ca. 3.7 days

captured in the model and, when we modelled and evaluated their performance, we found that for this supply chain, increasing ordering and delivery frequencies, reducing the producer's lead time and introducing new information systems improved chain performance. Our logical and empirical verifications demonstrated the model's validity. In future we intend to obtain generic building blocks in order to be able to model complex logistic systems in a more efficient way. We will start modelling larger supply chain problems generically, involving more links and/or activities.

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